

1. What is the purpose of using HumanMessage, AIMessage, and SystemMessage in LangChain? Explain the role of each message type.
2. Why is st.session_state required when building a chatbot in Streamlit? What problem does it solve?
3. What is a Streamlit rerun? Explain how reruns affect normal Python variables and why chat history would disappear without session state.
4. How does storing both HumanMessage and AIMessage help an AI chatbot maintain context during a conversation?
5. What is the purpose of a SystemMessage? Give two examples of instructions you could provide using a SystemMessage and explain how they influence the AI's responses.